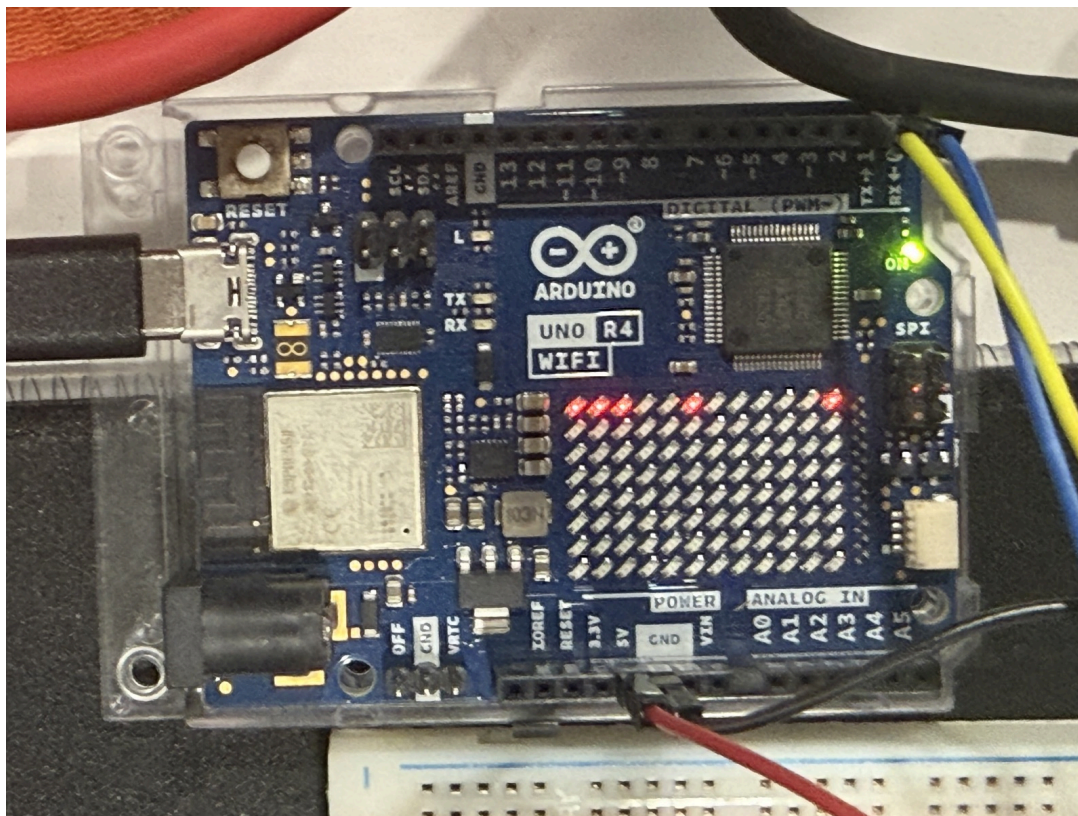


Reading the Uno LED display

The Uno R4 WiFi has a **12-column × 8-row red LED matrix**. It gives a quick view of connectivity and activity while you play. The green **ON** light and the small board **TX/RX** lights in the photograph are separate board LEDs, not pixels in the red matrix.

Evidence: the photograph below shows the owner's powered Uno on 2026-09-23. The meanings and examples on this page are **source-reviewed** against the current Uno firmware. A still photograph cannot establish which of the momentary activity or heartbeat pixels were on at a particular instant. The matrix animations and Uno audio response have not had a timed hardware acceptance test. [Matrix implementation](#), [hardware test log](#).



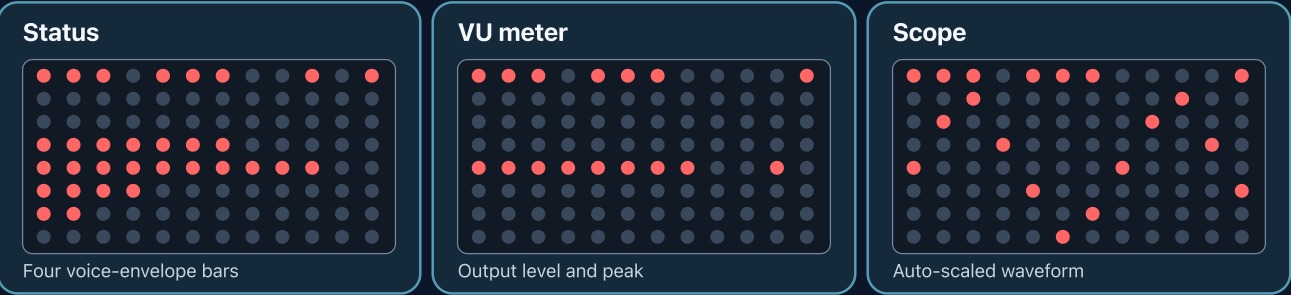
Live development build. The bright top-row pixels are a snapshot of indicators that can change within fractions of a second.

Choose a view

In the browser panel, open **Explore** or **All controls**, find **LED matrix**, and choose **Status**, **VU meter**, or **Scope**. These change only the Uno display; they do not change the sound. Status is the default after boot or a synth reset. The selector sends Pico-to-Uno Program Changes **10**, **11**, and **12** respectively. The Uno briefly scrolls the view name before returning to the selected display. These display commands are accepted only on the Pico UART, not on the separate DIN or RTP-MIDI inputs. [Browser control](#), [Uno display selection](#).

Uno R4 LED matrix views

Same 12x8 display · top row carries connection, activity, and heartbeat indicators



Illustrative frames, not captures. Red pixels are lit LEDs; positions change while the Uno runs.

Illustrative frames, not captures. Lit pixels vary with network traffic, notes, waveform, and the heartbeat.

View	What the lower rows show	Useful when
Status	Up to four horizontal bars, one for each synth voice. Each bar follows that voice's envelope level.	Checking which voices are active or releasing.
VU meter	One horizontal bar on the fifth row, plus a peak pixel. It follows the firmware's synthesized output level before the DAC.	Seeing relative output movement while playing.
Scope	A moving, automatically scaled trace with one dot per column. It samples the synthesized output before the DAC.	Seeing the shape and movement of the generated waveform.

VU and **Scope** are visual guides, not calibrated level meters or measurements at the audio jack. Because the audio output stage is not assembled yet, their real-world relationship to a mixer or headphones remains untested. [Output sampling and display calculations](#), [connection guide](#).

The top row stays useful in every view

Count columns **1–12 from left to right as viewed on the board**. The firmware draws these indicators on the top row in all three views. A pixel marked “brief” is expected to flash and go dark; it does not represent a persistent fault.

Uno R4 LED matrix · top row

Columns 1–12, left to right as viewed on the board · illustrative Status-view moment



Red = lit in this example · gray = unlit. Connection, activity, note, and heartbeat pixels may change.

- **1–6:** 1–3 Wi-Fi · 4 BLE-MIDI · 5 RTP-MIDI · 6 Pico link
- **7–12:** 7 Pico RX · 8 unused · 9 RTP RX · 10 note · 11 unused · 12 heartbeat

Column	Meaning in this build	When it lights
1–3	Uno Wi-Fi	Together while the Uno Wi-Fi stack reports connected.
4	BLE-MIDI	Never in this build; BLE-MIDI is disabled.
5	RTP-MIDI	While an RTP session reports connected.
6	Pico link	While the Uno has received a Pico readiness request within roughly 3.5 seconds. This is a handshake indicator, not merely a powered UART.
7	Pico UART receive, brief	For about 200 ms after a MIDI message arrives from the Pico.
8	Unused	Off.
9	RTP receive, brief	For about 300 ms after the Uno receives RTP MIDI activity.
10	Note flash in Status view	Briefly after a synth note starts. This flash is not drawn in VU or Scope.
11	Unused	Off.
12	Heartbeat	Toggles about every 250 ms; a regular blink shows the display loop is updating.

The **bottom-right pixel** is a timing diagnostic. It can light for a frame if more than 200 ms elapsed between LED updates, such as after a blocking operation. An occasional flash alone does not identify its cause. [Top-row and diagnostic overlays](#).

What happens at startup

The matrix first plays a boot animation and scrolls **QuarkWave**. While Wi-Fi starts, a two-pixel “comet” moves across the fourth row. The Uno then scrolls its IP address or **No WiFi**. An RTP connection can briefly scroll **RTP Connected**. When you change the display view, its name scrolls briefly; the regular matrix view resumes after the message hold interval. [Boot animation](#), [Wi-Fi display](#), [RTP notification](#).

Quick checks

- **Heartbeat blinks, Pico pixel is dark:** the display loop is running, but the Uno has not received a recent Pico readiness request. Check the browser's separate Pico and Uno connection labels and the inter-board wiring. The Pico-to-Uno link can be offline even while Uno Wi-Fi is connected.
- **Wi-Fi group is dark:** the Uno firmware does not currently report Wi-Fi connected. It can still play from its other available MIDI inputs, subject to their wiring.
- **Voice bars move but VU seems still:** switch to VU and play again. Status follows individual voice envelopes; VU follows the final synthesized output, so they are not identical.
- **No red pixels during a scrolling message:** give the scroll time to finish before judging the selected view.

These are reading aids, not a substitute for checking MIDI reception and audio on the assembled instrument. [Pico–Uno connection behavior](#), [hardware test log](#).